

Lecture 3 –Equilibration Protocols in MD Simulation

- Last time, we built three configurations using Packmol.
- Today, we will run all (L1P2, L2P2,L3P2) GROMACS simulations using these configurations.

Energy Minimization

- Energy minimization is performed to relax the initial geometry and reduce high potential energy
- A Steepest Descent Algorithm is used in GROMACS
- The Parameters we used are

No of steps: 50,000

Energy tolerance: $1000 \text{ kJ mol}^{-1} \text{ nm}^{-1}$

Cutoff distance: 1.2 nm

Time Step: 0.001 ps

Periodic boundary conditions: xyz

- Result

Stable minimized structure

Prepared for equilibration

NVT Equilibration

- It allows the system to stabilize temperature, relax the solvent and ions, and also maintain a fixed volume
- The parameters that we need are

Ensemble: NVT

Integrator: md

Time step: 0.001 ps

No of steps: 1,000,000 = 1ns

Temperature: 298 K

Thermostat: Velocity generation

Cutoff scheme: Verlet

Coulomb interaction: PME

Cutoff: 1.2 nm

NPT Equilibration

➤ NPT simulation is run to stabilize pressure, adjust system density, and volume. We use Parrinello–Rahman Barostat

➤ Parameters are

Time step: 0.001 ps

No of steps: 100,000,000 = 100ns

Temperature: 298 K

Thermostat: Nose–Hoover

Barostat: Parrinello–Rahman

Pressure: 1 bar

Cutoff: 1.2 nm

➤ Result

We will observe the stable density

Production Molecular Dynamics

- After the production run, we will collect trajectory data, Study ion transport and solvation behavior to analyze structural and dynamical properties
- We will use NPT
- Long simulation time
- Data recorded periodically
- Parameters used in the simulation are

Time step: 0.001ps

No of steps: 50000000 = 50ns

Trajectory saved every = 10 ps

- Result

Radial distribution functions

Diffusion coefficient

Ion coordination analysis

Thank you